# Finding Objective Function for Optimization

We have two Poses, $i$ and $j$.

$T_i$ = transform origin to $i$, $T_j$ = transform origin to $j$

- $T_i^{-1}$ is transformation st if we apply to world, world is now with origin at Pose $i$

To find $z_{ij}^{Prep}$, rel trans from $i$ to $j$ given poses:

We do $z_{ij} = T_i^{-1} \otimes T_j$ $\cdot \otimes$ Compositi

"Get into the shoes of Pose $i$, then go to Pose $j$"

We find this path

Now we also have the "true" tranformation which we store in our Pose Graph's edges. These are $z_{ij}$

We want to find the error (diff) $e_{ij}$ between these transformations to then minimize

$$e_{ij} = \hat{z}_{ij} \ominus z_{ij}^{pred}$$

$$= (\hat{z}_{ij})^{-1} \otimes z_{ij}^{pred}$$

- $$\hat{z}_{ij} = \begin{pmatrix} -\hat{x}\cos\hat{\theta} - \hat{y}\sin\hat{\theta} \\ \hat{x}\sin\hat{\theta} - \hat{y}\cos\hat{\theta} \\ -\hat{\theta} \end{pmatrix}$$

- $$z_{ij}^{pred} = \begin{pmatrix} (x_j - x_i)\cos\theta_i + (y_j - y_i)\sin\theta_i \\ -(x_j - x_i)\sin\theta_i + (y_j - y_i)\cos\theta_i \\ \theta_j - \theta_i \end{pmatrix}$$

$$= \begin{pmatrix} \Delta x \cos\theta + \Delta y \sin\theta_i \\ -\Delta x \sin\theta_i + \Delta y \cos\theta_i \end{pmatrix}$$

Given $a \otimes b = \begin{pmatrix} a_x + b_x \cos a_\theta - b_y \sin a_\theta \\ a_x + b_x \sin a_\theta + b_y \cos a_\theta \\ a_\theta + b_\theta \end{pmatrix}$

$$= \begin{pmatrix} (-\hat{x}\cos\hat{\theta} - \hat{y}\sin\hat{\theta}) + (-(x_j - x_i)\sin\theta_i + (y_j - y_i)\cos\theta_i)\cos(-\hat{\theta}) - (-(x_j - x_i)\sin\theta_i + (y_j - y_i)\cos\theta_i)\sin(-\hat{\theta}) \\ (-\hat{x}\cos\hat{\theta} - \hat{y}\sin\hat{\theta}) + (-(x_j - x_i)\sin\theta_i + (y_j - y_i)\cos\theta_i)\sin(-\hat{\theta}) + (-(x_j - x_i)\sin\theta_i + (y_j - y_i)\cos\theta_i)\cos(-\hat{\theta}) \\ -\hat{\theta} + (\theta_j - \theta_i) \end{pmatrix}$$

$\vdots$

$$= \begin{pmatrix} \\ \\ \theta_j - \theta_i - \hat{\theta} \end{pmatrix}$$